

Notation: $S = T_{kv} = KV$ context length

$$D = d_{model}, F = d_{FF}, N = n_q, K = n_{kv},$$

$$H = d_{head}, G = \frac{N}{K} = g = \text{query head groups per KV head}$$

input/output embed

$$1) \# \text{ params} = L(3DF + 2D(N+K)H + D) + 2DV = [(3DF + 4DNH + D) + 2DV]$$

$$= 64 \cdot (3 \cdot 4096 \cdot 16384 + 4 \cdot 4096^2 + 4096) + 2 \cdot 4096 \cdot 32000$$

$$\approx 11B$$

$$b) \# \text{ Attn params} = \frac{4DNH}{4DNH + 3D} = \frac{4D^2}{4D^2 + 12D} = \frac{4}{16} = \frac{1}{4}$$

$$c) \text{ Per token, the KV cache is } 2 \cdot L \cdot N \cdot H = 2LD = 2 \cdot 64 \cdot 4096 = 512KB$$

$$2) \text{ Total FLOPs} = 2BD^2, \text{ per device} = \frac{2BD^2}{xy} = \frac{BD^2}{16}$$

Because our computation isn't sharded across the Z dimension, we're doing Z extra FLOPs, meaning ~~2BD^2~~ $2BD^2Z$ FLOPs total.

$$3) \text{ Rule of thumb: Total maximal FLOPs} \approx 2 \times \text{product of every distinct dimension once} = 2IJKLMNO$$

$$4) AI = \frac{\text{FLOPs}}{\# \text{ bytes}} \quad \text{Self Attn: load QKV, compute softmax} \left(\frac{QK^T}{\sqrt{K}} \right) \cdot V, \text{ write result.}$$

• Total FLOPs as went through earlier = $4BTSNH$, ignoring softmax cost.

• #bytes: Read Q: ~~4BSNH~~ and write outputs $4BTSNH$

Read K, V = $4BSKH$

$$\text{Total} = 4BTSNH + 4BSKH$$

$$\therefore AI = \frac{4BTSNH}{4BTSNH + 4BSKH} = \frac{4TSN}{4TN + 4SK} \quad (\text{cancel } NKS)$$

$$= \frac{TSK}{TKG + 8K} \quad (\because N=KG)$$

$$= \frac{TS}{TG + 8}$$

During prefill $S=T$,

$$AI = \frac{T^2 G}{T(G+1)} = T \left(\frac{G}{G+1} \right) = O(1)$$

During generation, $T=1 \Rightarrow \frac{GS}{G+1}$, and at large context lengths S ,

$$AI \Rightarrow AI = \frac{GS}{G+1} = \frac{G}{\frac{1}{S} + 1} \rightarrow G \text{ as } S \rightarrow \infty$$

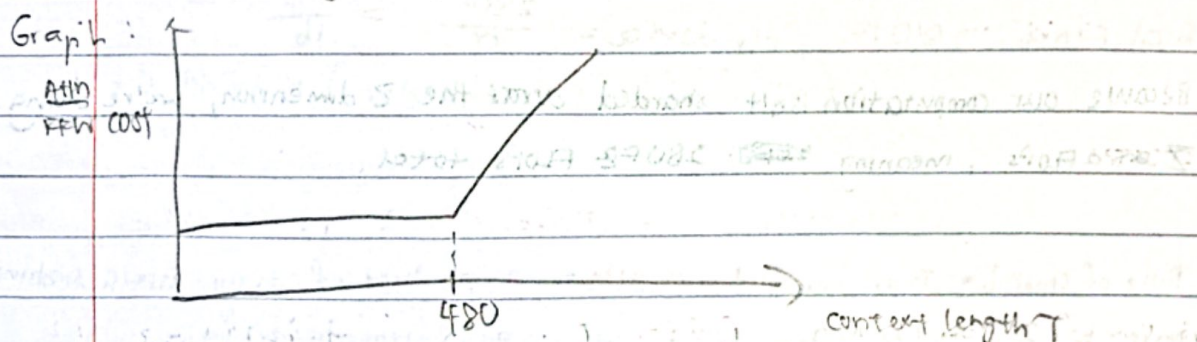
For TPU v5e, critical $AI \approx 240$,

Say for MHA, $AI = \frac{TG}{G+1} = \frac{T}{2} \because G=1 \text{ in MHA,}$

Then becomes compute bound at $\frac{T}{2} \geq 240 \Rightarrow T \geq 480$

For general GOA: $\frac{TG}{G+1} \geq 240 \Rightarrow T \geq 240 \left(\frac{G+1}{G} \right)$

Generation is rarely compute bound $\because$ one token per second



$$FFW \text{ FLOPs} = 6BTDF = 24BT^2, \quad SA \text{ FLOPs} = 4BT^2$$

$$\Rightarrow \frac{AI}{FFW} = \frac{T}{6D}$$

5) $SA \text{ FLOPs} \sim \text{OKVO prod FLOPs}$

$$\Rightarrow 24BT^2 \text{ NH} = 12BT^2 \text{ NH} \Rightarrow 2D = T \Rightarrow AT = 8192 \text{ if } D = 4096$$

One projection = $6BT^2 \text{ NH}$ $\because$ forward FLOPs = $2BT^2 \text{ NH}$

6) Must rematerialize: $QK^T = A$ & AV to get $\frac{\partial L}{\partial W_0}$

$$QK \rightarrow QK^T \rightarrow P = \text{softmax} \rightarrow Z = PV \rightarrow Y = ZW_0 \rightarrow \text{Loss}$$

To find ~~grad~~ grad of W_0 : $Y = ZW_0 \Rightarrow \frac{\partial L}{\partial W_0} = Z^T \frac{\partial L}{\partial Y} \Rightarrow Z = \sigma(QK^T)V$

$\Rightarrow$ must rematerialize $2 \cdot 4BT^2NH$

(because to recompute $QK^T: 2BT^2NH$, and $PV = 2BT^2NH$)

(Recording stopped here)

7)

(Chap 5 Note 2, strong scaling: $\uparrow$ # chips used for trng / inf & $\uparrow$ linearly throughput

① Data parallelism

③ Tensor parallelism

② fully-sharded data parallelism (FSDP)

④ pipeline parallelism

(Ignore this)

Ch4 Q7) 2.79M H800 hours

Theoretical: 3026 TFLOPs with sparsity $\Rightarrow$ 1513 TFLOPs/s w/o sparsity

$$\Rightarrow \text{Theoretical FLOPs achieved} = 1513e12 \times 2.79e6 \times 60^2 = 1.52e25$$

Actual: Rough estimate = GWT = $6(3713)(14.8e12) = 3.3e24$

$$\therefore \text{FLOPs util} = \frac{3.3e24}{1.52e25} = 21.7\%$$

393 INT8

8) To be compute bound, $T_{max} > T_{comm} \Rightarrow$ for TPU v5e, ~~1513~~ TFLOPs

~~Each~~ batch contains 8 tokens, and given that each token choose k

experts, assuming ~~each~~ tokens are spread evenly across E experts,

then # tokens per expert = $\frac{kB}{E}$. An expert matrix $x = W[D, F] \Rightarrow \text{FLOPs} =$

$2beDF$ (be = average # tokens processed by one expert) $\Rightarrow AI = \frac{2beDF}{DF} = 2be$

$= \frac{2kB}{E}$. From prev qns, TPU v5e becomes compute-bound at $AI = 240$

$$\Rightarrow \frac{2kB}{E} \geq 240 \Rightarrow B \geq \frac{120E}{k}$$

$$\text{DeepSeek: } E = 256, k = 8 \Rightarrow B \geq 3840 //$$